Gaussian Splatting for Physical Inverse Problems:

3D Gaussians for Inverse Scattering and 4D Gaussians for Fluid Dynamics

Thesis I Report

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Abstract

3D Gaussian Splatting represents a scene as a set of anisotropic Gaussians and has become a leading representation for photographic novel-view synthesis; its dynamic extension, 4D Gaussian Splatting, adds a deformation network over time. Both are designed for posed RGB images. This thesis asks whether the same representation is useful for *scientific inverse problems*, where an unknown physical field must be recovered from indirect, incomplete and noisy measurements produced by a known physics model. Using datasets and forward models from the InverseBench benchmark, we study two problems: (i) linear inverse scattering, where the permittivity of cells is recovered from scattered electromagnetic fields measured by 360, 180 or 60 receivers, in 2D and on 3D cell volumes; and (ii) the 2D incompressible Navier–Stokes equations, where an evolving vorticity field is reconstructed from dense or sparse observations over time with a 4D (space–time) Gaussian representation whose deformation is modelled either by a multilayer perceptron, as in 4D Gaussian Splatting, or by a Mamba selective state-space model. We replace the camera rasteriser with the physics operator, render Gaussians on a grid with a local-window evaluation, and train end-to-end by gradient descent with 3DGS-style densification and pruning.

The proof of concept, which includes a true 3D Born scattering model with sensors on a sphere, shows that (1) unbounded Gaussians lose to a regularised voxel grid in most settings, but rendering a bounded density $1 - e^{-f}$ (the opacity saturation of 3DGS) makes them beat the voxel grid under the true 3D model on every volume (up to +1.1 dB on real BBBC024 nuclei and +4.3 dB on synthetic nuclei) with about nine times fewer parameters; (2) adaptive densification can grow "ghost" copies of the object in regions the measurements cannot see (the operator's null space), which a simple sparsity term removes; and (3) for fluid flow, a deformation model is essential (+10 dB over static Gaussians), and a Mamba deformation model with temporal smoothness generalises to unobserved time steps better than the MLP of 4D Gaussian Splatting when only 10% of pixels are observed (25.4 vs. 23.0 dB). These results establish feasibility and point to the open problems addressed in Thesis II.

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Chapter 1

Introduction

1.1 Motivation

Many measurements in science do not show the quantity of interest directly. An optical tomography system records how light is scattered by a cell, not the cell's refractive index; a fluid experiment samples the flow at a few points and times, not the full velocity or vorticity field. Recovering the hidden quantity from such indirect, incomplete and noisy data is an *inverse problem*. Inverse problems are typically ill-posed: many different unknowns explain the measurements equally well, so the representation of the unknown and the prior knowledge built into it decide which solution is found.

In computer vision, 3D Gaussian Splatting (3DGS) [9] has recently changed how scenes are represented. Instead of a pixel or voxel grid, or a neural network evaluated at every point [16], a scene is a set of anisotropic Gaussians with learnable positions, shapes and amplitudes. The representation is compact, adapts its resolution to the content through densification and pruning, and is differentiable, so it can be fitted to measurements by gradient descent. Its dynamic extension, 4D Gaussian Splatting [27], keeps one canonical set of Gaussians and learns a deformation network that moves them over time.

Gaussians are also a natural basis for many physical fields: cell nuclei are smooth blobs, and vortices in a fluid are localised, smooth structures that are carried along by the flow. This suggests that a Gaussian representation could act as a useful structural prior for scientific inverse problems. Whether it does, and under which conditions, has not been studied systematically: existing Gaussian methods for scientific imaging target specific modalities such as X-ray computed tomography [2, 30], while the recent InverseBench benchmark [32] evaluates diffusion-model priors, not explicit Gaussian representations.

1.2 Problem statement (summary)

This thesis studies 3D and 4D Gaussian representations as the unknown in physics-based inverse problems. Given a known forward model $\mathcal{A}$ that maps a physical field to measurements, and measurements $\boldsymbol{y} = \mathcal{A}(f) + \boldsymbol{n}$, we represent f by Gaussians and fit their parameters so that the simulated measurements match $\boldsymbol{y}$. We ask when this representation helps compared with a plain grid of unknowns, how it fails, and how its motion over time should be modelled. The full problem definition and scope are given in Chapter 4.

1.3 Contributions of Thesis I

1. **Dataset analysis.** A study of the five InverseBench problems with respect to their suitability for 3D and 4D Gaussian representations, leading to the choice of linear inverse scattering (3D) and Navier–Stokes (4D).

2. **A physics-driven Gaussian pipeline.** A single implementation that renders 2D/3D Gaussians on a grid with a local-window evaluation, couples them to the InverseBench scattering operator (verified to match the benchmark implementation to within 3×10^{-7} relative error) and to observations of the Navier–Stokes solution, and trains them with densification, pruning and an optional sparsity term.

3. **4D Gaussians with a state-space motion model.** A 4D (2D + time) Gaussian representation whose deformation is modelled either by an MLP, as in 4D Gaussian Splatting, or by a Mamba selective state-space model [7] that treats each Gaussian's trajectory as a sequence.

4. **Initial results and a failure analysis.** Baseline comparisons against pixel/voxel optimisation with identical physics, and the identification of a "ghost" failure mode caused by the operator's null space, with a sparsity-based remedy.

5. **Open, reproducible code.** A public repository with a Python package, experiment configurations, notebooks, unit tests and all results so far (Appendix A).

1.4 Outline

Chapter 2 introduces inverse problems, the two physical models and Gaussian splatting. Chapter 3 reviews related work. Chapter 4 defines the problem, research questions and scope. Chapter 5 describes the pipeline and Chapter 6 its mathematical formulation. Chapter 7 gives the experimental setup and Chapter 8 the initial results. Chapter 9 discusses limitations and the plan for Thesis II.

Chapter 2

Background

2.1 Inverse problems

An inverse problem asks for an unknown f from measurements

$$\boldsymbol{y} = \mathcal{A}(f) + \boldsymbol{n}, \tag{2.1}$$

where $\mathcal{A}$ is a known forward model (here, a physics simulator) and $\boldsymbol{n}$ is noise. The problem is *ill-posed* when a solution does not exist, is not unique, or does not depend continuously on the data. In the problems studied here there are far fewer measurements than unknowns, so $\mathcal{A}$ has a large null space: many fields produce exactly the same measurements. A solution is therefore chosen by a prior, either explicit, as a regularisation term such as total variation [21] or sparsity, or implicit, through the *parametrisation* of the unknown. A convolutional network trained on a single image acts as such an implicit prior [24]; diffusion models trained on many examples act as a learned prior and are the subject of InverseBench [4, 32]. In this thesis the prior is the Gaussian representation itself, optionally combined with a sparsity term.

2.2 Linear inverse scattering

In optical diffraction tomography an object is illuminated by a known incident field and the scattered field is recorded by receivers around it [1]. For a weakly scattering object the total field inside the object can be approximated by the incident field (the first Born approximation), and the scattered field becomes a *linear* function of the object's permittivity contrast $f(\boldsymbol{x})$:

$$u_t^{\mathrm{sc}}(\boldsymbol{r}) = k_b^2 \int_\Omega G(\boldsymbol{r},\boldsymbol{x})\, f(\boldsymbol{x})\, u_t^{\mathrm{in}}(\boldsymbol{x})\, d\boldsymbol{x}, \qquad G(\boldsymbol{r},\boldsymbol{x}) = \tfrac{i}{4} H_0^{(1)}(k_b \|\boldsymbol{r}-\boldsymbol{x}\|)\,, \tag{2.2}$$

where t indexes the transmitter, $\boldsymbol{r}$ a receiver position, k_b the background wavenumber and $H_0^{(1)}$ the Hankel function of the first kind (the 2D Green's function). Multiple-scattering models drop the Born approximation at a higher cost [22]. InverseBench uses the linear model with 20 transmitters on a circle and 360, 180 or 60 receivers on the same circle, a 128×128 grid of a $0.18 \times 0.18\,\mathrm{m}$ domain, and cell images simulated with CytoPacq [26].

2.3 Incompressible Navier–Stokes flow

The 2D incompressible Navier–Stokes equations in vorticity form on the periodic domain $[0, 2\pi]^2$ are

$$\partial_t \omega + \boldsymbol{u} \cdot \nabla \omega = \frac{1}{Re} \Delta \omega + g, \qquad \boldsymbol{u} = (\partial_y \psi,\, -\partial_x \psi), \qquad -\Delta \psi = \omega, \tag{2.3}$$

with vorticity ω, velocity $\boldsymbol{u}$, stream function ψ, Reynolds number Re and forcing g. Vorticity is transported by the flow and diffused by viscosity, so its structures move and deform smoothly over time. InverseBench uses $Re = 200$, forcing $g = -4\cos(4y)$ and a pseudo-spectral solver; this problem is also a standard benchmark for neural PDE solvers [13].

2.4 Neural fields and Gaussian splatting

Neural radiance fields. NeRF [16] represents a scene by an MLP that maps a 3D position and viewing direction to colour and density and renders images by volume rendering along camera rays. Fourier features of the input coordinates allow MLPs to represent high frequencies [23]. Grid- and hash-based encodings make such fields much faster [17].

3D Gaussian Splatting. 3DGS [9] replaces the MLP by an explicit set of anisotropic 3D Gaussians, each with a mean, a covariance factorised into a rotation and scales, an opacity and a colour. Gaussians are projected ("splatted") onto the image plane and alpha-blended; all parameters are optimised by gradient descent. *Adaptive density control* clones or splits Gaussians where reconstruction gradients are large and prunes transparent ones, so the representation spends its capacity where the scene needs it.

Dynamic scenes. D-NeRF [18] introduced a canonical space plus a deformation network for dynamic NeRFs. 4D Gaussian Splatting [27] applies the same idea to Gaussians, predicting per-time offsets of position, scale and rotation from a spatio-temporal feature encoding (HexPlane [3]). Alternatives include per-frame Gaussians tracked over time [15] and native 4D Gaussians [28].

2.5 State-space sequence models and Mamba

Structured state-space models (S4) [8] model sequences with a linear recurrence $\boldsymbol{h}_t = \bar{\mathbf{A}}\boldsymbol{h}_{t-1} + \bar{\mathbf{B}}x_t$, $y_t = \mathbf{C}\boldsymbol{h}_t$, obtained by discretising a continuous-time system. Mamba [7] makes the step size Δ and the matrices $\mathbf{B}$, $\mathbf{C}$ functions of the input (*selection*), so the model can decide what to keep in its state, and computes the recurrence with a hardware-efficient scan in time linear in the sequence length. Mamba has been applied to vision backbones [33]. Here it is used to model how each Gaussian moves through time.

Chapter 3

Literature Review

This chapter reviews four lines of work that meet in this thesis: (1) learned priors for scientific inverse problems, (2) explicit Gaussian representations, (3) Gaussian and neural-field methods for scientific and medical imaging, and (4) dynamic representations and sequence models. It closes with the gap this thesis addresses.

3.1 Priors for scientific inverse problems

Classical solvers combine a data-fidelity term with hand-crafted regularisers such as total variation [21]. Deep learning added two families of priors. *Untrained* priors use the structure of a network as the regulariser: Deep Image Prior [24] fits a randomly initialised convolutional network to a single corrupted image and stops early. *Learned* priors are trained on many examples; diffusion models are currently the most general, and Diffusion Posterior Sampling [4] guides the reverse diffusion with the gradient of the data likelihood, so the same prior can serve many forward models.

InverseBench [32] evaluates 14 such plug-and-play diffusion-prior algorithms on five scientific problems: linear inverse scattering, compressed-sensing MRI on fastMRI [29], black hole imaging, full waveform inversion on OpenFWI [5], and the 2D Navier–Stokes equations. Its findings that motivate this thesis are that (i) scientific forward models differ strongly from image-restoration operators (complex-valued, non-linear, expensive, or without gradients), (ii) diffusion priors help most when measurements are sparse, and (iii) they can fail when outputs violate physical constraints. All its methods represent the unknown on a pixel grid. Physics-informed neural networks [20] and neural operators such as FNO [13] are complementary: they learn solution operators of PDEs rather than priors over unknowns.

3.2 Explicit Gaussian representations

3DGS [9] showed that an explicit, adaptive set of anisotropic Gaussians can match or exceed neural fields [16, 17] in quality while rendering in real time. Its key ingredients are a differentiable rasteriser, a covariance parametrisation by rotation and scale that stays positive semi-definite, and adaptive density control. GaussianImage [31] transfers the idea to 2D: an image is represented by 2D Gaussians and rendered by accumulated summation, achieving fast decoding and compression. Our 2D and 3D representations follow the same principle, but are evaluated on a regular grid and supervised through a physics operator instead of pixels.

3.3 Gaussians and neural fields in scientific imaging

Tomography. X-Gaussian [2] adapts Gaussian splatting to X-ray projection imaging with a radiative Gaussian model, and R^2-Gaussian [30] rectifies the integration bias of splatting for tomographic volume reconstruction, showing that Gaussians can serve as a voxel-free representation of attenuation volumes. Both rely on ray-based projection, the natural analogue of camera rendering; scattering measurements are instead a wave-based integral over the whole object.

Astronomy. BH-NeRF [11] reconstructs 3D emission around a black hole over time with a neural field constrained by gravitational lensing and orbital dynamics, an example of physics replacing multi-view geometry.

Microscopy data. Simulated cell images with exact ground truth are produced by CytoPacq and related generators [26]; the Broad Bioimage Benchmark Collection [14] distributes such synthetic 3D nucleus datasets (BBBC024), which we use as real 3D cell shapes.

3.4 Dynamic representations and sequence models

D-NeRF [18] and 4D Gaussian Splatting [27] model motion with a canonical representation and a deformation field queried independently at each time; HexPlane [3] and K-Planes [6] factorise space–time features into planes. Dynamic 3D Gaussians [15] track persistent Gaussians frame by frame, and native 4D Gaussians [28] slice 4D primitives in time. Language-embedded variants (LangSplat [19], 4D LangSplat [12]) attach semantic features to such Gaussians, the starting point of this thesis' original direction. None of these treats a Gaussian's trajectory as a sequence. State-space models [7, 8] are efficient sequence

Table 3.1: Representative related work. "Meas." is the kind of measurement used for supervision.

Work	Representation	Meas.	Relation to this thesis
NeRF [16]	MLP field	RGB images	implicit field; rendering by rays
3DGS [9]	3D Gaussians	RGB images	our representation, without cameras
4DGS [27]	Gaussians + deformation	video	our 4D design (MLP deformation)
GaussianImage [31]	2D Gaussians	pixels	2D Gaussian rendering on a grid
X-Gaussian [2], R^2-Gaussian [30]	3D Gaussians	X-ray projections	Gaussians for physics, ray-based
BH-NeRF [11]	neural field	radio interferometry	physics replaces multi-view geometry
DPS [4], InverseBench [32]	pixel grid + diffusion prior	physics operators	same problems, learned prior
Mamba [7]	selective SSM	sequences	our motion model over time
This thesis	2D/3D/4D Gaussians	scattering; sparse flow samples	Gaussians + physics; Mamba motion

models that could do exactly that, and have been adopted in vision [33]; their use as a motion model for 4D Gaussians is, to our knowledge, unexplored.

3.5 Synthesis and research gap

Table 3.1 summarises the landscape. Diffusion-prior methods are general but represent the unknown on a grid and need large training sets; Gaussian methods are compact and adaptive but have been demonstrated mainly with camera- or ray-based measurements. The gap addressed here is the use of 3D/4D Gaussian representations as the unknown in *wave-based and PDE-based scientific inverse problems*, the analysis of their failure modes under severe ill-posedness, and sequence-model motion priors for their dynamics.

Chapter 4

Problem Definition and Research Scope

4.1 Problem definition

Let $f : \Omega \to \mathbb{R}$ be an unknown physical field on a domain $\Omega \subset \mathbb{R}^d$ (permittivity contrast of a cell, $d \in \{2, 3\}$), or a time-dependent field $\omega : \Omega \times [0, T] \to \mathbb{R}$ (vorticity of a flow, $d = 2$). A known forward model $\mathcal{A}$ produces measurements $\boldsymbol{y} = \mathcal{A}(f) + \boldsymbol{n}$ with $\dim \boldsymbol{y} \ll \dim f$. We represent the unknown by a set of N Gaussians with parameters $\Theta = \{\theta_i\}_{i=1}^{N}$ (and, for dynamic fields, a deformation network with weights ϕ) and solve

$$\hat{\Theta} = \arg\min_{\Theta} \; \mathcal{D}\Big(\mathcal{A}(f_\Theta), \boldsymbol{y}\Big) + \mathcal{R}(\Theta), \tag{4.1}$$

where $\mathcal{D}$ measures data misfit and $\mathcal{R}$ is an optional regulariser. The ground truth is never used for fitting; it is used only to evaluate the reconstruction.

4.2 Research questions

RQ1 Can 2D/3D Gaussian representations recover physical fields from scattering measurements, and how do they compare with a pixel/voxel grid optimised with the same physics as the number of measurements decreases?

RQ2 What failure modes arise when an adaptive Gaussian representation meets an operator with a large null space, and how can they be prevented?

RQ3 Can a 4D Gaussian representation (canonical Gaussians plus deformation) reconstruct an evolving fluid field from dense and sparse observations, including at time steps that were never observed?

RQ4 Does a selective state-space model (Mamba) over each Gaussian's trajectory model the dynamics better than the per-time MLP deformation of 4D Gaussian Splatting?

4.3 Scope

In scope (Thesis I).

- Linear inverse scattering with the InverseBench forward model: 2D cells (InverseBench test set) and 3D cell volumes (synthetic HL60-like nuclei and CytoPacq/BBBC024 nuclei), with 360, 180 and 60 receivers and the benchmark noise level.
- 3D physics with two models: the 2D operator applied to every z-slice (*slice-wise*) and a *true 3D* first-order Born model with transmitters and receivers on a sphere around the volume.
- 2D Navier–Stokes ($Re = 200$) as a 2D + time ("4D") problem: sequences of 21 frames, observations of all or 10% of the pixels per frame, every fourth frame held out.
- Baselines: direct optimisation of a pixel/voxel grid with the same data term (and the same sparsity term).
- Metrics: PSNR, SSIM, relative ℓ_2 error, number of Gaussians, run time.

Out of scope (deferred to Thesis II).

- A full 3D scattering model with sensors on a sphere, and non-linear (multiple-scattering) physics.
- Recovering the Navier–Stokes *initial condition* through the solver, as posed in InverseBench.
- Exhaustive benchmarking against the diffusion-prior methods of InverseBench; training learned priors.
- Real (non-simulated) measurement data, and the remaining InverseBench problems.

4.4 Assumptions

The forward models are known exactly and are differentiable; measurement noise is additive Gaussian at the benchmark level ($\sigma = 10^{-4}$ for scattering); permittivity contrast is non-negative and bounded in $[0, 1]$; vorticity is periodic in space.

Chapter 5

Proposed Methodology and Pipeline Architecture

5.1 Overview

The same training loop serves every task (Figure 5.1). A set of Gaussians is rendered on the grid of the unknown; a physics operator turns the rendered field into simulated measurements; the misfit to the real measurements, plus an optional sparsity term, is back-propagated to all Gaussian parameters; and an adaptive density control step periodically adds and removes Gaussians. For dynamic fields a deformation network maps the canonical Gaussians to their state at each time step before rendering.

5.2 Gaussian field representation

Each Gaussian has a centre, per-axis scales, an orientation and an amplitude; the field at a grid point is the sum of all Gaussians evaluated there (Section 6.1). Because a Gaussian is negligible beyond about three standard deviations, each Gaussian is evaluated only in a window of $\pm h$ cells around its centre and accumulated into the grid with a scatter-add, so the cost is $O(N(2h+1)^d)$ and independent of the grid size; scales are clamped to $h/3$ so that the window always covers $\pm 3\sigma$. For permittivity the amplitude is kept non-negative with a softplus; for vorticity it is signed and the window wraps around the periodic domain. All parameters of one scene form a single $N \times P$ tensor ($P = 6$ in 2D, $P = 11$ in 3D), which is convenient for sequence models.

5.3 Physics operators

Scattering. We re-implement the InverseBench linear scattering operator as two complex matrix products in single precision on the GPU and verify at start-up that it matches

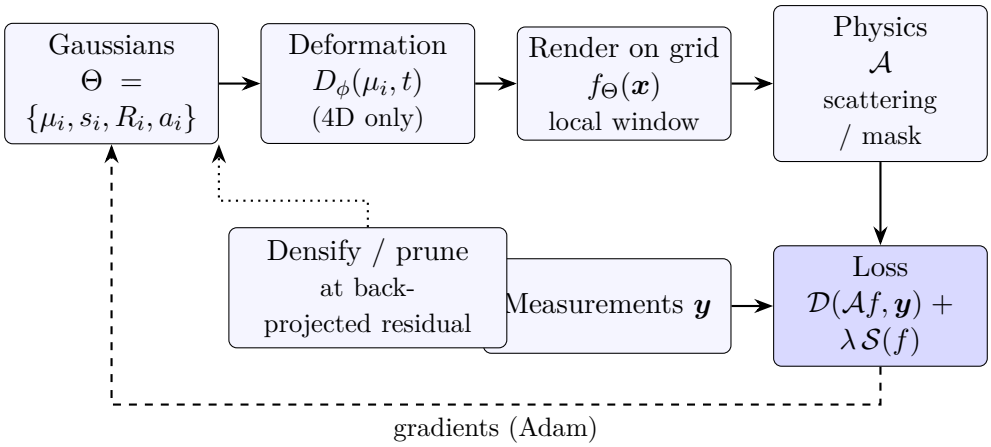


Figure 5.1: Pipeline. The camera rasteriser of 3DGS is replaced by rendering on a grid followed by the physics operator.

the benchmark implementation to within a relative error of 3×10^{-7}. For 3D volumes we use either the slice-wise operator (applied to every z-slice, losses summed) or the true 3D Born operator (6.4), implemented the same way with precomputed incident fields and receiver Green's functions ($P \times T$ and $P \times M$ complex matrices; 1.5 GB for 360 receivers).

Bounded Gaussians. In the final configuration the Gaussian field is passed through the saturation (6.2) before the physics operator, so it obeys the same $[0, 1]$ range as the grid baseline, and it can be initialised from the grid reconstruction (options `-bounded` and `-init_from_grid`).

Fluid observations. For Navier–Stokes the measurement operator selects the observed pixels of each frame (a random mask with 100% or 10% coverage); frames that are held out for evaluation are never observed.

5.4 Training and adaptive density control

All parameters are optimised with Adam [10] and a cosine learning-rate schedule. For scattering, Gaussians are initialised uniformly at random, with no knowledge of the object. Every 250 iterations, during the first 60% of training, new Gaussians are inserted at the grid points with the largest magnitude of the *back-projected residual* $\operatorname{Re}\mathcal{A}^{\mathsf{H}}(\boldsymbol{y} - \mathcal{A}f)$, i.e. where the measurements indicate missing structure, and Gaussians whose amplitude has fallen below 10^{-3} are pruned. This replaces the image-space gradient criterion of 3DGS, which is unavailable without pixel supervision.

5.5 Sparsity regulariser

Early experiments showed that aggressive densification can place mass in regions to which the measurements are blind (Section 8.2). A sparsity term on the total intensity of the rendered field penalises mass the data does not require. Because the field is non-negative, the mean of the field equals its ℓ_1 norm divided by the grid size.

5.6 4D representation of fluid flow

A canonical set of 2D Gaussians is initialised from the observed pixels of the first frame. A deformation network maps each Gaussian's canonical centre and the time t to offsets of all its parameters; positions move in pixel units. Three deformation models are compared:

- **MLP** (as in 4D Gaussian Splatting): Fourier features of (μ_i, t) fed to a three-layer MLP, evaluated independently for every Gaussian and time.
- **Mamba**: for each Gaussian the inputs at all T time steps form a sequence processed by two Mamba blocks, so the offset at time t depends on the Gaussian's earlier states (causal).
- **Bidirectional Mamba**: a second Mamba stack reads the sequence backwards and the two are summed, so every time step sees both past and future.

The output layer is initialised to zero so training starts from the canonical configuration. An optional temporal smoothness term penalises differences between offsets at consecutive times, including unobserved times. A *static* model without deformation serves as an ablation.

5.7 Baselines

The baseline for scattering optimises every pixel (2D) or voxel (3D) directly with the same data term, starting from zero and projected to $[0, 1]$ after each step; optionally it uses the same sparsity term. Only the representation differs between the two methods. For fluids, the static-Gaussian model is the baseline.

5.8 Implementation

The pipeline is a Python/PyTorch package (`thesis_gs`) with modules for the Gaussian field, physics, volume loading, scattering experiments, Navier–Stokes data, deformation networks and 4D training, a command-line interface, experiment configurations and unit tests (Appendix A). Mamba is implemented in pure PyTorch with an explicit selective scan, which is sufficient for sequences of 21 steps and runs on any GPU.

Chapter 6

Mathematical Formulation

6.1 Gaussian field

A field on a grid with points $\boldsymbol{x} \in \Omega \subset \mathbb{R}^d$ is represented by N Gaussians,

$$f_\Theta(\boldsymbol{x}) = \sum_{i=1}^{N} a_i \exp\Big(-\tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\Big), \qquad \boldsymbol{\Sigma}_i = \mathbf{R}_i \operatorname{diag}(\boldsymbol{s}_i)^2 \mathbf{R}_i^\top, \tag{6.1}$$

with centre $\boldsymbol{\mu}_i \in \mathbb{R}^d$, scales $\boldsymbol{s}_i = \exp(\boldsymbol{\ell}_i)$ clamped to $[s_{\min}, h/3]$, and rotation $\mathbf{R}_i$: in 2D $\mathbf{R}(\vartheta_i) = \left(\begin{smallmatrix} \cos\vartheta_i & -\sin\vartheta_i \\ \sin\vartheta_i & \cos\vartheta_i \end{smallmatrix}\right)$, in 3D the rotation matrix of the normalised quaternion $\boldsymbol{q}_i/\|\boldsymbol{q}_i\|$. The factorisation keeps $\boldsymbol{\Sigma}_i$ symmetric positive definite for any parameter values [9]. The amplitude is $a_i = \operatorname{softplus}(\alpha_i) \geq 0$ for permittivity and $a_i = \alpha_i \in \mathbb{R}$ for vorticity. The parameters per Gaussian are $\theta_i = (\boldsymbol{\mu}_i, \boldsymbol{\ell}_i, \vartheta_i \text{ or } \boldsymbol{q}_i, \alpha_i)$. In practice the sum in (6.1) runs only over Gaussians whose window $\|\boldsymbol{x} - \operatorname{round}(\boldsymbol{\mu}_i)\|_\infty \leq h$ contains $\boldsymbol{x}$. A *generalised* Gaussian replaces the quadratic form $q_i(\boldsymbol{x})$ in the exponent by $q_i(\boldsymbol{x})^\beta$ with $\beta \geq 1$: $\beta = 1$ is (6.1), larger β gives a flatter top and a sharper edge, intended for piecewise-constant objects.

Bounded density. The sum (6.1) is unbounded: overlapping Gaussians can exceed the physical range of the contrast, whereas the grid baseline is projected onto $[0, 1]$ after every step. The bounded variant renders

$$\tilde{f}_\Theta(\boldsymbol{x}) = 1 - \exp\Big(- f_\Theta(\boldsymbol{x})\Big) \in [0, 1), \tag{6.2}$$

the same saturation as opacity accumulation in 3DGS. It is smooth, monotone and close to f_Θ for small values, so it changes only where Gaussians overlap. Optionally the Gaussians are *initialised from the grid solution*: centres are sampled in proportion to $-\log(1 - \min(f_{\text{grid}}, 0.95))$, the density that (6.2) maps back to the grid reconstruction, and then refined with the same physics loss.

6.2 Scattering forward model

On a grid of P points with spacing $\Delta x\,\Delta y$, the linear (Born) scattering operator of the benchmark maps a contrast image $\boldsymbol{f} \in \mathbb{R}^P$ to measurements for T transmitters and M receivers,

$$y_{t,r} = \Delta x\,\Delta y \sum_{p=1}^{P} f_p\,\overline{u_t^{\mathrm{in}}(\boldsymbol{x}_p)}\,g_r(\boldsymbol{x}_p), \qquad g_r(\boldsymbol{x}) = k_b^2\,\tfrac{i}{4}H_0^{(1)}\big(k_b\|\boldsymbol{r}_r - \boldsymbol{x}\|\big), \tag{6.3}$$

written compactly as $\boldsymbol{y} = \mathbf{A}\boldsymbol{f} + \boldsymbol{n}$ with $\mathbf{A} \in \mathbb{C}^{TM\times P}$ and complex Gaussian noise of standard deviation $\sigma = 10^{-4}$. Here $T = 20$, $M \in \{360, 180, 60\}$ and $P = 128^2$, so $TM \le 7200$ complex measurements describe 16384 unknowns and $\mathbf{A}$ has a non-trivial null space. For a volume with slices $\boldsymbol{f}^{(z)}$, $z = 1, \dots, Z$, the slice-wise model is $\boldsymbol{y}^{(z)} = \mathbf{A}\boldsymbol{f}^{(z)} + \boldsymbol{n}^{(z)}$.

True 3D Born model. The slice-wise model treats each slice as an independent 2D experiment. A true 3D first-order Born model with point transmitters at $\boldsymbol{s}_t$ and receivers at $\boldsymbol{r}_r$ on a sphere around the volume is

$$y_{t,r} = k^2\,\Delta V \sum_{p=1}^{P} f_p\,G(\|\boldsymbol{s}_t - \boldsymbol{x}_p\|)\,G(\|\boldsymbol{r}_r - \boldsymbol{x}_p\|), \qquad G(d) = \frac{e^{\mathrm{i}kd}}{4\pi d}, \tag{6.4}$$

with the free-space Green's function of the 3D Helmholtz equation, voxel volume $\Delta V = \Delta x^3$ and the same voxel size, wavelength (six voxels) and sensor radius (1.6 m) as the 2D benchmark. We use $T = 60$ transmitters and $M \in \{360, 180, 60\}$ receivers spread evenly over the sphere (golden-angle spiral). A $32 \times 128 \times 128$ volume has $P = 524{,}288$ unknowns but only $TM \le 21{,}600$ complex measurements, about a tenth of the slice-wise model, so the problem is far more under-determined. Noise is complex Gaussian with standard deviation 2% of the RMS measurement, the signal-to-noise ratio of the 2D benchmark. The implementation matches the analytic response of a point scatterer to 10^{-7} and passes an adjoint test $\langle \mathbf{A}\boldsymbol{f}, \boldsymbol{r}\rangle = \langle \boldsymbol{f}, \mathbf{A}^H\boldsymbol{r}\rangle$.

6.3 Objective for scattering

With $\boldsymbol{f}_\Theta$ the field (6.1) sampled on the grid,

$$\mathcal{L}(\Theta) = \frac{\sum_z \left\|\mathbf{A}\boldsymbol{f}_\Theta^{(z)} - \boldsymbol{y}^{(z)}\right\|_2^2}{\sum_z \|\boldsymbol{y}^{(z)}\|_2^2} \;+\; \lambda\,\frac{1}{|\Omega|}\sum_{\boldsymbol{x}\in\Omega} f_\Theta(\boldsymbol{x}), \tag{6.5}$$

where the second term equals $\lambda\,\|\boldsymbol{f}_\Theta\|_1 / |\Omega|$ because $f_\Theta \ge 0$. The baseline solves the same problem over the grid values directly, $\min_{\boldsymbol{f}\in[0,1]^{|\Omega|}} \|\mathbf{A}\boldsymbol{f} - \boldsymbol{y}\|_2^2 / \|\boldsymbol{y}\|_2^2 + \lambda\,\overline{\boldsymbol{f}}$, by projected Adam from $\boldsymbol{f} = \mathbf{0}$.

Why grids do not grow ghosts. The gradient of the data term with respect to $\boldsymbol{f}$ is $2\,\mathrm{Re}\,\mathbf{A}^{\mathsf{H}}(\mathbf{A}\boldsymbol{f}-\boldsymbol{y})/\left\|\boldsymbol{y}\right\|^2$, which lies in the range of $\mathbf{A}^{\mathsf{H}}$, the orthogonal complement of the null space. Gradient steps from zero therefore never add null-space components (up to the effect of the projection onto $[0,1]$). Gaussians break this property: a Gaussian inserted anywhere carries mass that may lie largely in the null space, where the data term exerts no force to remove it. The sparsity term in (6.5) supplies that force.

6.4 Adaptive density control

Every $K = 250$ iterations while $k \le 0.6\,K_{\max}$ the back-projected residual

$$\boldsymbol{\rho} = \mathrm{Re}\Big(\mathbf{A}^{\mathsf{H}}(\boldsymbol{y} - \mathbf{A}\boldsymbol{f}_{\Theta})\Big) \tag{6.6}$$

is computed, n_{add} new Gaussians are created at the grid points with the largest $|\rho|$ (scale 1.5 cells, amplitude proportional to $|\rho|$), and Gaussians with $a_i < 10^{-3}$ are removed.

6.5 4D representation

For a sequence $\omega_0, \dots, \omega_{T-1}$ observed through masks $\mathbf{M}_t$, canonical parameters θ_i are deformed by a network D_ϕ:

$$\theta_i(t) = \theta_i + \mathbf{S}\,D_\phi\Big(\gamma(\boldsymbol{\mu}_i/L),\,\gamma(t/T)\Big), \qquad \gamma(p) = \Big[\sin(2^k\pi p), \cos(2^k\pi p)\Big]_{k=0}^{K-1}, \tag{6.7}$$

where $\mathbf{S}$ scales position offsets to pixels and γ are Fourier features [23]. The training loss over a random batch $\mathcal{B}$ of observed frames is

$$\mathcal{L}(\Theta,\phi) = \sum_{t\in\mathcal{B}} \frac{\left\|\mathbf{M}_t \odot (f_{\Theta(t)} - \tilde{\omega}_t)\right\|_2^2}{\left\|\mathbf{M}_t\right\|_1} \;+\; \lambda_T \sum_{t\in\mathcal{B}} \frac{1}{N}\sum_i \left\|D_\phi(\cdot, t+1) - D_\phi(\cdot,t)\right\|_2^2, \tag{6.8}$$

with $\tilde{\omega}_t$ the (normalised) observations and $\lambda_T \ge 0$ the temporal smoothness weight.

6.6 Mamba deformation

For Gaussian i, the inputs $\boldsymbol{x}_t$ ($t = 0, \dots, T-1$) are linear embeddings of $[\gamma(\boldsymbol{\mu}_i), \gamma(t)]$. A Mamba block [7] computes, after a gated input projection and a short causal convolution,

$$\Delta_t = \mathrm{softplus}(\mathbf{W}_\Delta \boldsymbol{x}_t), \qquad \mathbf{B}_t = \mathbf{W}_B \boldsymbol{x}_t, \qquad \mathbf{C}_t = \mathbf{W}_C \boldsymbol{x}_t, \tag{6.9}$$

$$\boldsymbol{h}_t = \exp(\Delta_t \mathbf{A}) \odot \boldsymbol{h}_{t-1} + \Delta_t\,\mathbf{B}_t\,\boldsymbol{x}_t, \qquad \boldsymbol{y}_t = \mathbf{C}_t \boldsymbol{h}_t + \mathbf{D}\boldsymbol{x}_t, \tag{6.10}$$

with a diagonal state matrix $\mathbf{A} = -\exp(\mathbf{A}_{\log})$, followed by multiplicative gating with $\mathrm{SiLU}(\boldsymbol{z}_t)$ and a residual connection. Because $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ depend on the input, the model selects what to remember along the trajectory. The bidirectional variant adds a second stack applied to the time-reversed sequence and sums the two outputs.

6.7 Evaluation metrics

For a reconstruction $\hat{f}$ and ground truth f with data range R,

$$\mathrm{PSNR} = 10 \log_{10} \frac{R^2}{\frac{1}{|\Omega|} \left\| \hat{f} - f \right\|_2^2}, \qquad \mathrm{relL2} = \frac{\left\| \hat{f} - f \right\|_2}{\| f \|_2}, \tag{6.11}$$

and SSIM [25] computed with the default 7×7 uniform window of scikit-image ($R = 1$ for permittivity, $R = 2$ for normalised vorticity). For 4D, metrics are averaged separately over observed (training) and held-out frames.

Chapter 7

Experimental Setup

7.1 Data

- **2D cells (scattering).** The first samples of the InverseBench `inv-scatter-test` set: 128×128 permittivity images of 1–6 HL60 nuclei simulated with CytoPacq [26], values in $[0, 1]$.
- **3D cells (scattering).** (a) Synthetic HL60-like nuclei generated by our code: 1–6 soft, rotated ellipsoids with smooth internal texture in a $32 \times 128 \times 128$ volume. (b) Real CytoPacq nuclei from BBBC024 of the Broad Bioimage Benchmark Collection [14] (ground-truth label masks, 20 nuclei per volume), converted to foreground masks, cropped to the central half in x and y and resampled to $32 \times 128 \times 128$.
- **Fluid (4D).** Initial vorticity fields from the InverseBench `navier-stokes-test` set ($Re = 200$), rolled forward with the InverseBench pseudo-spectral solver to 21 frames spaced $\Delta t = 0.05$ apart, divided by 10 to lie in about $[-1, 1]$. Frames 2, 6, 10, 14 and 18 are held out; the rest are observed completely or at a random 10% of pixels per frame.

7.2 Forward models and noise

Scattering uses the InverseBench configuration: domain $0.18 \times 0.18\,\mathrm{m}$, wavelength six grid cells, 20 transmitters and 360, 180 or 60 receivers on a circle of radius 1.6 m, complex Gaussian noise $\sigma = 10^{-4}$ with a fixed seed per sample so that Gaussian and baseline fits see identical measurements.

Table 7.1: Default settings (overridable from the command line; every run stores its exact settings).

Setting	Scattering 2D / 3D	Fluid 4D
Iterations	3000	3000
Optimiser, learning rate	Adam, 0.05 (Gaussians), 0.01 (grid)	Adam, 0.01 (Gaussians), 10^{-3} (network)
Schedule	cosine	cosine
Max. / initial Gaussians	2000 / 250 (2D); 8000 / 2000 (3D)	2000 / 2000
Window half-size h	8 (2D), 6 (3D)	10, periodic
Densification	every 250 it., until 60%, $+n_{\text{init}}/4$	none
Sparsity λ	0, 0.1, 0.3, 1, 3	–
Temporal smoothness λ_T	–	0 (initial), 1 (final)
Deformation	–	MLP (3×128), Mamba and bidirectional Mamba (2 blocks, $d = 64$, state 16)
Batch	one sample	4 frames
Samples per setting	3 (5 in the final 2D runs)	3 sequences

7.3 Hyper-parameters

7.4 Hardware and software

All experiments ran on a single NVIDIA Tesla T4 (16 GB) on Google Colab or Kaggle, with Python 3, PyTorch, NumPy, SciPy and scikit-image. A 2D scattering fit takes about 13 s; a 3D fit a few minutes; a 4D fit with the MLP about 3 min and with Mamba 15–28 min (sequential scan; the longer time with temporal smoothness).

7.5 Protocol

For each sample, receiver count and method, measurements are generated once and both the Gaussian and the grid method are fitted to them. Reported numbers are means over samples; per-sample values, figures and run configurations are stored in the repository (`results/`).

Chapter 8

Initial Baseline Results

Numbers are means over samples (PSNR in dB / SSIM); the number of samples is given in each table. The Gaussian and grid methods always use identical measurements and data terms. Initial runs (October 2026) used three samples; the final runs (November 2026) used five and then ten samples for 2D scattering, added a true 3D scattering model, and added the experiments marked as final.

8.1 Representation check

Fitting Gaussians directly to the ground truth (no physics) confirms that the representation can express the fields: 48.2 dB / 0.998 on the 2D cells with about 680 Gaussians, and 65.2 dB / 0.999 on the synthetic 3D volumes with 5500 Gaussians.

8.2 2D inverse scattering on InverseBench cells

Table 8.1 lists the reconstructions from measurements only. Without regularisation, Gaussians are 6–7 dB below the pixel grid with 360 or 180 receivers and 2.7 dB below with 60. Increasing the Gaussian budget did *not* help: with aggressive densification (about 2000 Gaussians) PSNR fell to 22.9 / 22.6 / 19.6 dB. The reconstructions show why (Figure 8.1): a faint copy of the cell appears outside the object. These "ghosts" sit where the back-projected residual is large but the measurements carry almost no information, so the data term cannot remove them (Chapter 6). Adding the sparsity term removes the ghosts. With $\lambda = 0.3$ Gaussians improve at every receiver count, and at 60 receivers they exceed the unregularised pixel grid in SSIM (0.924 vs. 0.886 over five samples; 0.940 vs. 0.878 over the first three) while staying 1–1.5 dB lower in PSNR. Larger weights ($\lambda = 1, 3$) shrink the cells and lower both metrics; $\lambda = 0.1$ gives slightly higher PSNR than 0.3 at 360 and 180 receivers.

The final runs add a *fair* baseline: the pixel grid with the same sparsity term. It reaches

27.7 dB / 0.933 at 60 receivers and exceeds the Gaussians in both metrics at every receiver count. The SSIM advantage at 60 receivers therefore comes from the sparsity prior, not from the Gaussian representation itself. Sparsity is, however, *necessary* for Gaussians (it prevents ghosts) while it costs the grid PSNR at high receiver counts (36.1 → 33.2 dB at 360).

A run on ten cells confirms this with paired comparisons: with the same sparsity term, the pixel grid is better than the Gaussians on *every* cell and at every receiver count, by 0.6 ± 0.3, 2.3 ± 0.7 and 2.9 ± 0.8 dB at 60, 180 and 360 receivers. The spread between cells (standard deviation about 3–4 dB) is larger than most differences between methods, which is why paired comparisons are reported. Generalised Gaussians with $\beta = 2$ lower PSNR by about 1–2 dB. Against the *unregularised* grid, Gaussians with sparsity still have higher SSIM at 60 receivers on eight of ten cells (0.951 vs. 0.910).

Table 8.1: 2D inverse scattering, InverseBench cells (PSNR dB / SSIM).

	Method	360 receivers	180 receivers	60 receivers
Initial (3 samples)	Gaussians	30.7 / 0.959	30.4 / 0.957	26.5 / 0.814
	Gaussians, ∼2000 (aggressive densification)	22.9 / –	22.6 / –	19.6 / –
	Gaussians + sparsity, $\lambda = 0.3$	31.4 / 0.983	31.0 / 0.982	28.2 / 0.940
	Gaussians + sparsity, $\lambda = 1$	27.6 / 0.958	27.6 / 0.958	25.2 / 0.908
	Gaussians + sparsity, $\lambda = 3$	21.6 / 0.875	21.6 / 0.876	20.5 / 0.826
	Pixel grid	36.7 / 0.985	35.3 / 0.976	29.2 / 0.878
Final (5 samples)	Gaussians + sparsity, $\lambda = 0.1$	31.8 / 0.982	31.5 / 0.980	27.4 / 0.909
	Gaussians + sparsity, $\lambda = 0.3$	30.8 / 0.980	30.6 / 0.979	27.1 / 0.924
	Pixel grid	**36.1** / 0.984	**34.7** / 0.974	**28.6** / 0.886
	Pixel grid + sparsity, $\lambda = 0.3$	33.2 / **0.985**	32.5 / **0.983**	27.7 / **0.933**
Final (10 samples)	Gaussians + sparsity, $\lambda = 0.3$	32.2 / 0.985	31.9 / 0.984	29.4 / 0.951
	Generalised Gaussians ($\beta = 2$) + sparsity	31.4 / 0.982	31.1 / 0.981	27.7 / 0.934
	Pixel grid	**37.3** / 0.986	**35.9** / 0.976	**30.7** / 0.910
	Pixel grid + sparsity, $\lambda = 0.3$	35.0 / **0.990**	34.2 / **0.988**	30.0 / **0.957**

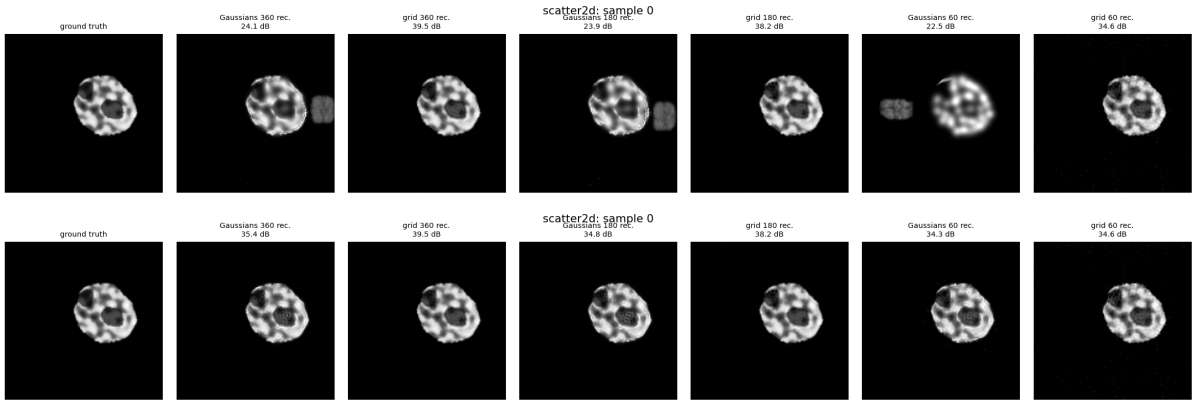


Figure 8.1: Cell 0. Top: aggressive densification without regularisation produces a ghost copy beside the cell. Bottom: with the sparsity term ($\lambda = 0.3$) the ghost is gone. Columns: ground truth, then Gaussian and pixel reconstructions at 360, 180 and 60 receivers.

8.3 3D inverse scattering, slice-wise model

On synthetic nuclei (Table 8.2, Figure 8.2) Gaussians are almost unaffected by the number of receivers (40.6–40.9 dB), while the voxel grid degrades from 44.1 to 37.3 dB and becomes speckled. With 60 receivers Gaussians are better on all three volumes (+3.6 dB on average) and have higher SSIM at every receiver count.

The real CytoPacq nuclei from BBBC024 reverse this picture. The voxel grid reaches 35–42 dB, while Gaussians stay at 27–28 dB regardless of the number of receivers or the sparsity term. Figure 8.3 shows the reason: the BBBC024 volumes are binary masks with sharp nucleus boundaries, and a sum of smooth Gaussians reproduces each nucleus with a soft edge and a faint bright rim. The error is concentrated at the boundaries and is a representation limit (the PSNR does not change with the receiver count), not a measurement limit. In SSIM, which is less sensitive to edge error, Gaussians with sparsity (0.955) still beat the unregularised voxel grid (0.936) at 60 receivers, but not the voxel grid with the same sparsity (0.971). Together with the synthetic result this gives a clear rule: Gaussians help when the object is smooth, and hurt when it is piecewise constant with sharp edges. Sharper generalised Gaussians ($\beta = 4$) did not fix this: although they fit binary masks about 3 dB better when fitted directly to the ground truth, under the physics loss they reach 28.1, 28.0 and 25.5 dB, slightly below ordinary Gaussians.

Table 8.2: 3D inverse scattering, $32 \times 128 \times 128$ volumes (PSNR dB / SSIM, mean of 3 volumes).

Volumes	Method	360 receivers	180 receivers	60 receivers
Synthetic nuclei	Gaussians	40.8 / **0.994**	40.6 / **0.994**	**40.9** / **0.988**
	Voxel grid	**44.1** / 0.987	**41.5** / 0.977	37.3 / 0.932
BBBC024 (CytoPacq)	Gaussians	27.9 / 0.967	27.6 / 0.965	27.0 / 0.943
	Voxel grid	**41.8** / 0.989	**40.3** / 0.980	**35.3** / 0.936
	Gaussians + sparsity, $\lambda = 0.3$	28.4 / 0.974	28.4 / 0.974	27.4 / 0.955
	Voxel grid + sparsity, $\lambda = 0.3$	38.5 / **0.996**	38.0 / **0.995**	34.4 / **0.971**
	Gen. Gaussians ($\beta = 4$) + sparsity	28.1 / 0.973	28.0 / 0.973	25.5 / 0.932

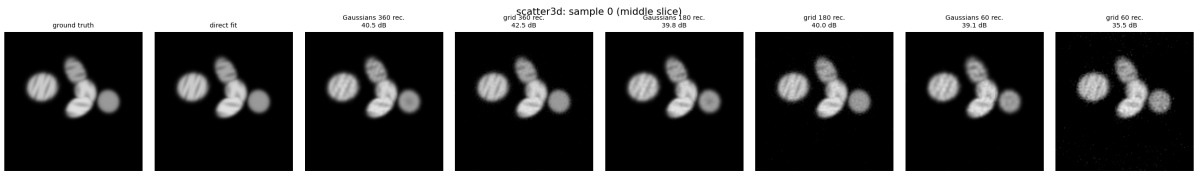


Figure 8.2: Middle slice of synthetic volume 0: ground truth, direct fit, then Gaussian and voxel reconstructions at 360, 180 and 60 receivers.

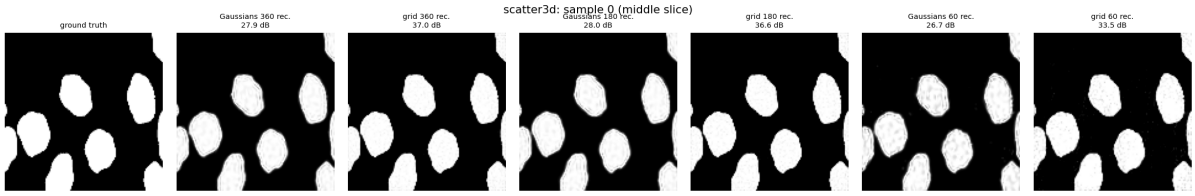


Figure 8.3: Middle slice of BBBC024 volume 0 (sparsity $\lambda = 0.3$): ground truth, then Gaussian and voxel reconstructions at 360, 180 and 60 receivers. Gaussians soften the sharp nucleus boundaries and leave a faint rim.

8.4 True 3D inverse scattering

Table 8.3 and Figure 8.4 report the true 3D Born model (6.4) with sensors on a sphere, sparsity $\lambda = 0.3$ for both methods. With about ten times fewer measurements than the slice-wise model, all numbers are lower, and the voxel grid is better than the Gaussians in every setting: by 4.5–6 dB on synthetic nuclei and by 4.6–7.0 dB on BBBC024 (better on every volume except one synthetic volume at 180 receivers). The robustness of Gaussians to fewer receivers seen with the slice-wise model does not carry over: both methods lose about 5–7 dB from 360 to 60 receivers (Section 8.5 shows how bounding the density reverses this). Generalised Gaussians ($\beta = 4$) are again worse (24.2 / 22.6 / 16.3 dB on BBBC024).

The reconstructions show where Gaussians lose. On BBBC024 they place a bright rim at each nucleus boundary and leave the interior below the true value, while the voxel grid, which is projected onto $[0, 1]$ after every step, fills the nuclei uniformly. On binary objects this box constraint is a strong prior that the Gaussians lack: their summed density is unbounded during optimisation. The Gaussians are, however, about **11–12 times more compact**: 3700–4600 Gaussians (41k–51k numbers) against 524,288 voxels.

Table 8.3: True 3D inverse scattering (3D Born model, 60 transmitters, receivers on a sphere), $32 \times 128 \times 128$ volumes, sparsity $\lambda = 0.3$ for both methods (PSNR dB / SSIM, mean of 3 volumes).

Volumes	Method	Parameters	360 receivers	180 receivers	60 receivers
Synthetic nuclei	Gaussians	≈42k	37.6 / 0.977	36.6 / 0.955	32.1 / 0.852
	Voxel grid	524k	**42.1 / 0.985**	**39.9 / 0.976**	**38.2 / 0.966**
BBBC024 (CytoPacq)	Gaussians	≈47k	25.6 / 0.960	24.9 / 0.940	20.9 / 0.772
	Gen. Gaussians ($\beta = 4$)	≈51k	24.2 / 0.954	22.6 / 0.927	16.3 / 0.722
	Voxel grid	524k	**32.5 / 0.983**	**30.5 / 0.956**	**25.6 / 0.875**

8.5 Bounded Gaussians close the gap

The analysis above pointed to one asymmetry: the grid is bounded to $[0, 1]$, the Gaussian sum is not. Rendering the bounded density (6.2) instead, with everything else unchanged,

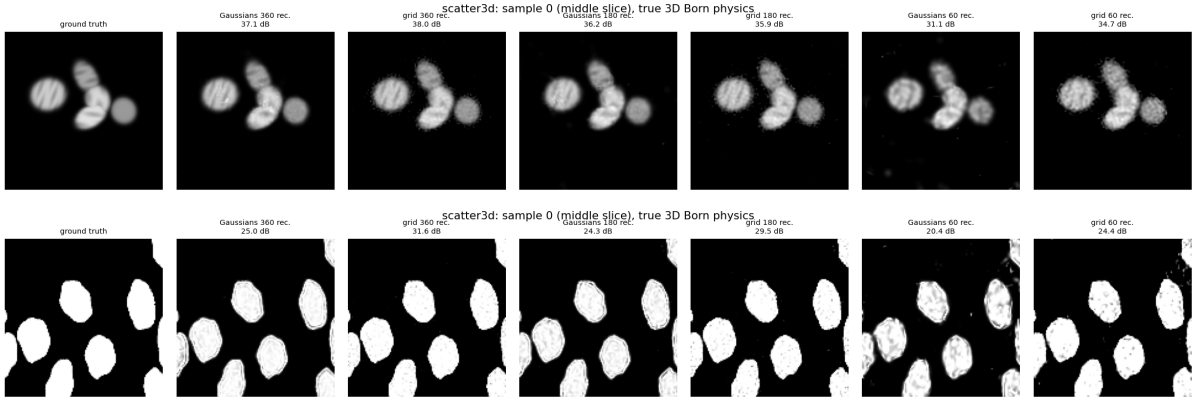


Figure 8.4: True 3D Born model, middle slices of volume 0. Top: synthetic nuclei. Bottom: BBBC024. Columns: ground truth, then Gaussian and voxel reconstructions at 360, 180 and 60 receivers. Gaussians leave bright rims at the boundaries of the binary nuclei.

reverses the 3D results (Table 8.4, Figure 8.5). Under the true 3D model on BBBC024, bounded Gaussians gain 6–7 dB and now match the voxel grid with the same sparsity term at 360 receivers and beat it at 180 and 60 receivers; initialised from the grid solution they beat it on *every* volume at every receiver count (paired differences $+0.26 \pm 0.07$, $+0.97 \pm 0.09$ and $+1.12 \pm 0.10$ dB at 360, 180 and 60 receivers) with higher SSIM, while using about 60k parameters instead of 524k. On synthetic nuclei the gain is larger: +4.3, +3.7 and +0.9 dB over the voxel grid, again better on every volume. The rims and under-filled interiors of Figure 8.4 disappear.

The gain is smaller where the grid is already well conditioned. With the slice-wise model (ten times more measurements) bounded, grid-initialised Gaussians improve from 28.4 to 35.8 dB on BBBC024 but remain below the regularised voxel grid at 360 and 180 receivers (38.5, 38.0 dB) and tie at 60 (34.1 vs. 34.4 dB, higher SSIM). On textured 2D cells bounded Gaussians beat the regularised pixel grid at 60 receivers on six of ten cells (+0.4 dB on average, within the spread) and trail it by 0.2–1.4 dB at 180 and 360 receivers. The pattern is consistent: *the fewer the measurements relative to unknowns, the more the compact, bounded Gaussian representation helps.*

8.6 4D Gaussians for Navier–Stokes

Table 8.5 reports PSNR on held-out frames, with observed-frame PSNR in brackets. Static Gaussians reach only about 17 dB: the flow changes too much over 21 frames for one fixed field. A deformation network adds about 10 dB. Without regularisation, Mamba fits the observed frames much better than the MLP (36.1 vs. 30.3 dB with full observations), but on held-out frames the two are within 0.3 dB: Mamba's extra capacity is spent memorising observed times.

Table 8.4: Bounded Gaussians (B) and bounded Gaussians initialised from the grid (B+G) vs. the grid with the same sparsity term ($\lambda = 0.3$). PSNR dB / SSIM; parameters in brackets.

Setting	Method	Params	360 receivers	180 receivers	60 receivers
True 3D, BBBC024 (3)	Gaussians (unbounded)	47k	25.6 / 0.960	24.9 / 0.940	20.9 / 0.772
	Gaussians B	47k	32.5 / **0.991**	31.3 / **0.983**	**26.8** / 0.903
	Gaussians B+G	60k	**32.8** / 0.990	**31.4** / 0.980	26.7 / **0.909**
	Voxel grid	524k	32.5 / 0.983	30.5 / 0.956	25.6 / 0.875
True 3D, synthetic (3)	Gaussians B+G	54k	**46.4 / 0.993**	**43.6 / 0.986**	**39.1** / 0.963
	Voxel grid	524k	42.1 / 0.985	39.9 / 0.976	38.2 / **0.966**
Slice-wise, BBBC024 (3)	Gaussians B+G	60k	35.8 / 0.995	35.7 / 0.995	34.1 / **0.983**
	Voxel grid	524k	**38.5 / 0.996**	**38.0** / 0.995	**34.4** / 0.971
2D cells (10)	Gaussians B	3.2k	33.7 / 0.988	33.3 / 0.988	30.4 / 0.957
	Gaussians B+G	4.0k	34.4 / 0.989	34.0 / **0.989**	**30.5** / 0.957
	Pixel grid	16.4k	**35.0 / 0.990**	**34.2** / 0.988	30.0 / 0.957

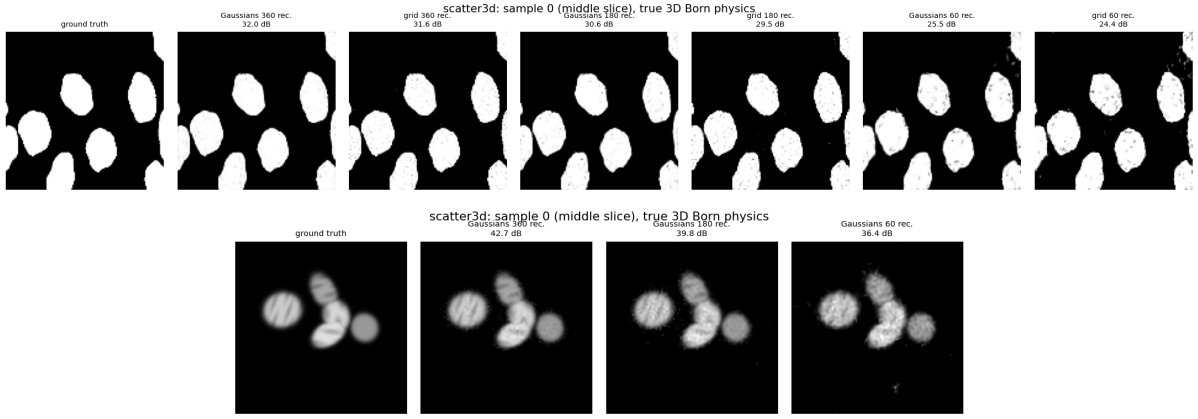


Figure 8.5: True 3D model with bounded, grid-initialised Gaussians. Top: BBBC024 volume 0, ground truth, then Gaussians and voxel grid at 360, 180 and 60 receivers (compare Figure 8.4). Bottom: synthetic volume 0, ground truth and Gaussians at 360, 180 and 60 receivers (voxel grid: 38.0 / 35.9 / 34.7 dB in Figure 8.4).

The final runs add the temporal smoothness term ($\lambda_T = 1$). With 10% of pixels observed, Mamba now generalises clearly better than the MLP: 25.4 vs. 23.0 dB on held-out frames, better on each of the three sequences (+2.2, +2.0 and +2.9 dB), and 2.3 dB better than unregularised Mamba. The MLP is unchanged by the term, and the MLP reconstructions show streak artefacts where data are sparse (Figure 8.7). With full observations all deformation models are within about 1 dB (MLP 27.4, Mamba 27.2 dB). The bidirectional Mamba fits observed frames best but does not improve held-out frames (25.1 and 26.4 dB) and doubles the training time. Mamba is about nine times slower than the MLP (28 vs. 3 min per sequence) because of its sequential pure-PyTorch scan.

Table 8.5: 4D Navier–Stokes, PSNR (dB) on held-out frames; observed-frame PSNR in brackets (mean of 3 sequences). "+ smooth" = temporal smoothness $\lambda_T = 1$ (final runs).

Observed pixels	Static	MLP (4DGS-style)	Mamba	Bidirectional Mamba
100%	17.0 (16.8)	27.6 (30.3)	**27.9** (36.1)	–
100% + smooth	–	27.4 (30.6)	27.2 (35.2)	26.4 (**36.5**)
10%	16.0 (15.9)	23.2 (24.0)	23.1 (27.7)	–
10% + smooth	–	23.0 (24.1)	**25.4** (28.9)	25.1 (**29.6**)
Training time per sequence	3 min	3 min	15–28 min	52 min

8.7 Answers to the research questions

RQ1 Yes in the most under-determined setting, once the density is bounded. Under the true 3D model, bounded Gaussians beat a voxel grid with the same sparsity prior on every volume (up to +1.1 dB on real BBBC024 nuclei and +4.3 dB on synthetic nuclei) with about nine times fewer parameters. With better-conditioned measurements (slice-wise 3D, 2D at 360 and 180 receivers) the regularised grid remains slightly better. Unbounded Gaussians lose in almost every setting: bounding the density is essential.

RQ2 Adaptive densification can populate the operator's null space with ghost structures; a sparsity term on total intensity prevents them and is required for Gaussians.

RQ3 Yes: canonical Gaussians plus a deformation network reconstruct the flow at held-out times, about 10 dB better than static Gaussians, from full or 10% observations.

RQ4 Yes, when observations are sparse and the model is regularised in time: Mamba with temporal smoothness beats the MLP by 2.4 dB on held-out frames with 10% observations. With full observations there is no gain, and the bidirectional variant adds cost without benefit.

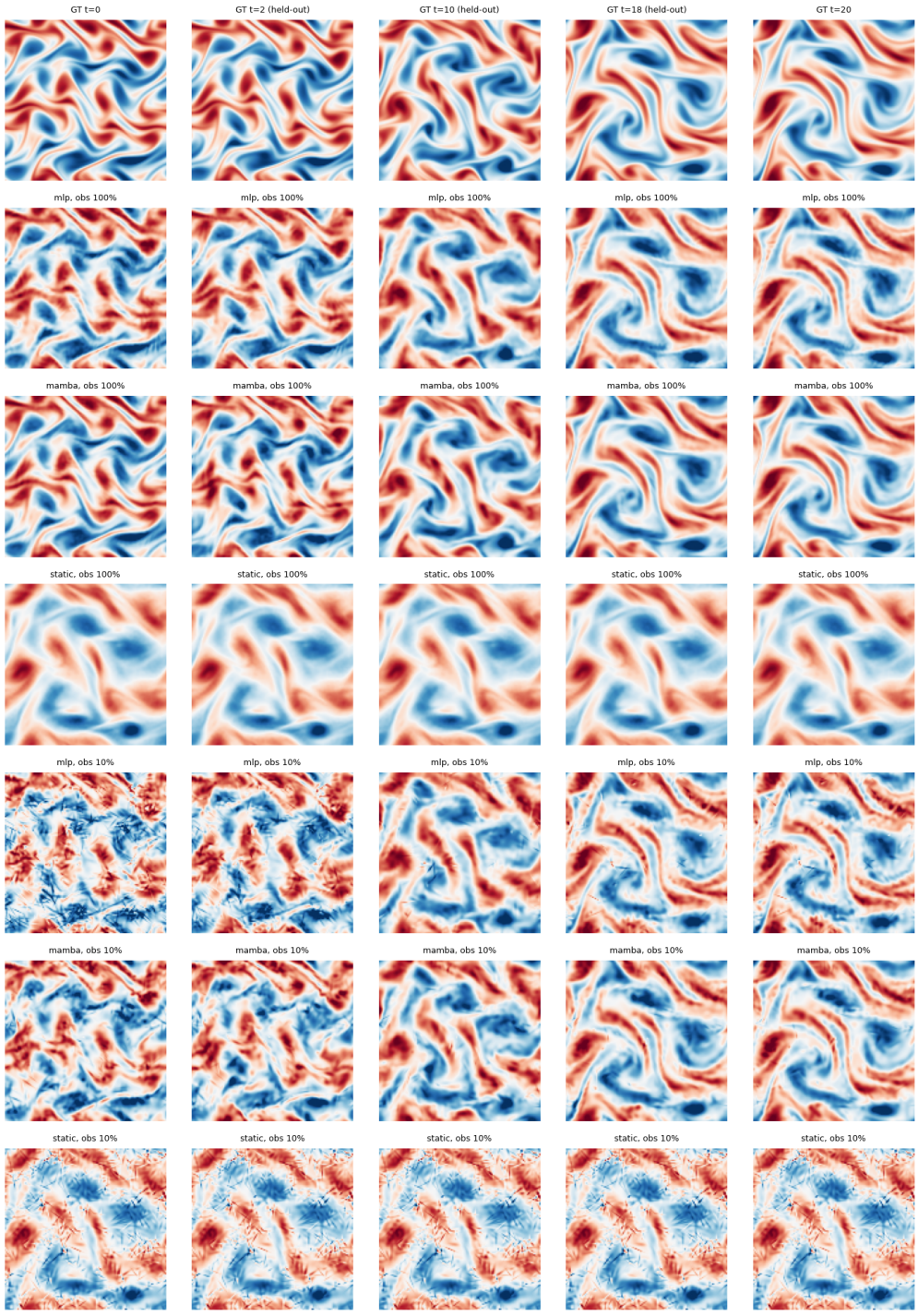


Figure 8.6: Initial runs: vorticity of sequence 0 at five times (held-out frames marked). Rows: ground truth, then MLP, Mamba and static models with 100% and 10% of pixels observed.

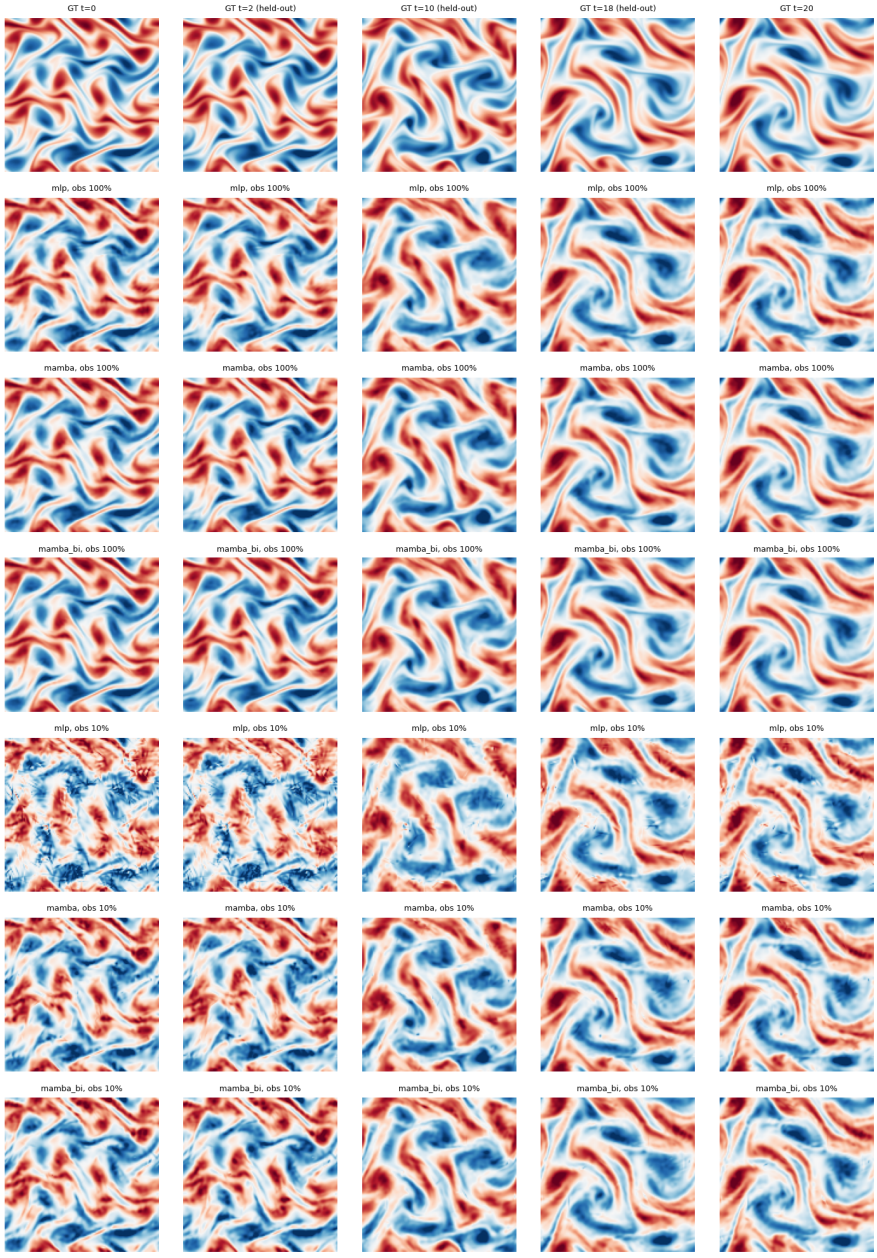


Figure 8.7: Final runs with temporal smoothness: ground truth, then MLP, Mamba and bidirectional Mamba with 100% and 10% of pixels observed. With 10% observations the MLP shows streaks; Mamba stays coherent.

Chapter 9

Discussion, Limitations and Plan for Thesis II

9.1 Discussion

Replacing the camera model of Gaussian splatting with a physics operator works: the same representation, optimiser and density control recover cells from scattered fields and moving vorticity from sparse samples. The representation acts as a prior that is helpful when the object matches it (smooth nuclei, scarce data) and neutral or harmful when it does not (fine texture, abundant data). The most instructive finding is the ghost failure mode: adaptivity, the main strength of Gaussian splatting with cameras, becomes a liability when the measurements are blind to part of the domain, and must be balanced by an explicit prior. The final runs add a second lesson: a Gaussian sum is unbounded, and on physical quantities with a known range this matters more than the shape of the kernel. Unbounded Gaussians lost to a clipped grid almost everywhere; the bounded density $1 - e^{-f}$, the same saturation as opacity in 3DGS, turned the true 3D results into a consistent win with nine times fewer parameters, while sharper kernels did not help. For dynamics, the deformation model matters far more than the choice of network; a sequence model pays off only once regularised in time, and then mainly when observations are sparse.

9.2 Limitations

- The true 3D model is linear (first Born approximation); multiple scattering is not modelled, and 3D results use three volumes.
- Results use three to five samples per setting; differences of a few tenths of a dB are not yet significant.

- The 4D problem reconstructs a field from observations of itself; InverseBench's harder task of recovering the initial condition through the solver is not addressed.
- No learned prior is used, and no comparison with InverseBench's diffusion methods is made (by design for Thesis I).
- Hyper-parameters (sparsity weight, Gaussian budget, window size, smoothness weight) were tuned on few samples.
- Under better-conditioned measurements (slice-wise 3D, 2D with many receivers) bounded Gaussians remain slightly below a regularised grid.

9.3 Plan for Thesis II

1. Multiple scattering in 3D, and better optimisation of Gaussians under the true 3D model (initialisation from a back-projection or grid solution, bounded densities).
2. Systematic evaluation on the full InverseBench test sets with confidence intervals, and comparison with the benchmark's diffusion-prior methods.
3. Physics-aware density control (densify only within the support indicated by the data; null-space-aware regularisers) instead of a global sparsity term.
4. Bounded Gaussians on the full test sets with confidence intervals, and learned or adaptive saturation for quantities without a fixed range.
5. Motion models: hardware-efficient Mamba (parallel scan), tuning of the smoothness weight, and advection-informed deformation that uses the velocity field implied by the vorticity.
6. The Navier–Stokes initial-value inverse problem through a differentiable solver.
7. Connection to the original direction of this work: attaching semantic features to physical Gaussians, in the spirit of LangSplat [19] and 4D LangSplat [12].

9.4 Conclusion

This Thesis I establishes a working, reproducible proof of concept for 3D and 4D Gaussian representations in two scientific inverse problems, identifies when they help and how they fail, and defines the experiments that Thesis II will scale up.

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Appendix A

Reproducibility

The code, configurations and results are in the thesis repository (`src/thesis_gs` package, `bin/` scripts, `configs/`, `notebooks/`, `results/`, `tests/`). InverseBench and its test data are downloaded automatically; BBBC024 is downloaded by the Kaggle notebook.

```
# install, smoke test, 2D run, final runs, unit tests
pip install -r requirements.txt
python bin/thesis-gs --task all --quick --out outputs/smoke
python bin/thesis-gs --task scatter2d --l1 0.3 --out outputs/s2d
python bin/run_experiments.py configs/final_part_a.json \
       --out outputs/final_a --volumes /path/to/BBBC024
python bin/run_experiments.py configs/final_part_b.json \
       --out outputs/final_b
pytest -q
```

Every run writes `run_config.json` with its exact settings, `results.csv` with per-sample metrics, `summary.csv` with means, and `reconstructions.png`.